

BIGGS TREE GROUPS

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ABSTRACT. Biggs gave an explicit construction, using finite colored trees, of finite permutation groups whose Cayley graphs have valence C and girth tending to infinity as the radius R of the tree tends to infinity. We show that when the number of colors is at least 3, the group so presented contains the full alternating group on the vertices of the tree.

This gives, for each $C \geq 3$, an infinite family of pairs $(G_{C,R}, S_{C,R})$ such that $G_{C,R}$ is an alternating or symmetric group, $S_{C,R}$ is a generating set of $G_{C,R}$ of size C with an explicit permutation description of its generators, and such that the sequence of Cayley graphs $\text{Cay}(G_{C,R}, S_{C,R})$ has constant valence C and girth tending to infinity as R tends to infinity.

1. INTRODUCTION

Let C be a finite set of colors, and let Γ be a finite simplicial graph such that each edge is assigned a color from C , and such that no two edges incident to a common vertex have the same color (a *proper coloring*). We may assume every color of C is actually used on some edge. Then each $c \in C$ determines a nontrivial permutation of the vertices of Γ by swapping the two vertices of each edge of color c . Let G be the permutation group generated by C .

Biggs [4] calls the colored graph a ‘picture’ and the group G the ‘story of the picture’. See also the survey [5]. Later works, eg [2], refer to G as a ‘Biggs group’, although this is a bit of a misnomer, as it is the choice of generating set that is novel, not the group itself.

By construction, each $c \in C$ is an involution, since it is a product of disjoint transpositions, so the Cayley graph $\text{Cay}(G, C)$ of G with respect to the generating set C is a C -valent¹, undirected graph. Of particular interest is the case that $\Gamma = T_{C,R}$ is the closed ball of radius R centered at a root vertex o of the C -regular C -colored tree T_C . The number of vertices of $T_{C,R}$ is $N_{C,R} := \frac{C(C-1)^R - 2}{C-2}$ when $C > 2$. By numbering the vertices $1, \dots, N_{C,R}$, in whatever way, we may consider the permutation group $G_{C,R}$ as a subgroup of $\text{Sym}(N_{C,R})$. For $C = 3$, Biggs [3] showed that $\text{girth}(\text{Cay}(G_{C,R}, C)) \geq 2R + 1$. This was the first example of a family of cubic (3-valent) Cayley graphs with unbounded girth. The girth argument also generalizes to $C \geq 3$ [7, Theorem 19].

It is not hard to see ([5, Lemma 7.1] or Proposition 3.6) that $G_{2,R}$ is a finite dihedral group. Exoo and Jajcay [8] note that in small examples where $G_{C,R}$ can be explicitly computed, it turns out that when $C > 2$ the group $G_{C,R}$ is the full symmetric group or the alternating group. However, $|\text{Sym}(N_{C,R})|$ grows quickly, and only for the very smallest examples can the isomorphism type of $G_{C,R}$ be checked by computer. The purpose of this paper is to explain the experimental observation of Exoo and Jajcay. Our main tasks are to use the colored tree $T_{C,R}$ to show that $G_{C,R}$ is 2-transitive, hence primitive, and to produce some cycles. Then

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¹Whenever C appears where an integer is expected, interpret the notation to mean $|C|$.

we can apply a theorem (Theorem 2.1) of Jones that uses the Classification of Finite Simple Groups to refine the Jordan Symmetric Group Theorem. This leads quickly to the conclusion that either $G_{C,R}$ contains the alternating group or it is a subgroup of some projective semi-linear group of the same permutation degree. We rule out the latter using a variety of methods, applying to different ranges of C and R :

- (1) Demonstrate the existence of cycles with many fixed points.
- (2) Show there is no projective semi-linear group of the same degree.
- (3) Show that $G_{C,R}$ contains an element whose order is too high relative to the permutation degree for it to occur in a projective semi-linear group.

The conclusion is the main theorem of this paper, whose proof occupies Section 5.

Main Theorem. *For $C \geq 3$:*

$$G_{C,R} \cong \begin{cases} \text{Alt}(N_{C,R}) & \text{if } C \text{ and } R \text{ are even} \\ \text{Sym}(N_{C,R}) & \text{otherwise} \end{cases}$$

Figure 1 shows $\text{Cay}(G_{C,R}, C)$ for $C = \{r, g, b\}$ and $R = 1$ embedded in a torus, obtained by identifying opposite sides of the dashed hexagon. It is a Cayley graph of $\text{Sym}(4)$ that is isomorphic to the Nauru graph. The order of $G_{3,2}$ is already $10! = 3628800$, so we will not try to draw its Cayley graph.

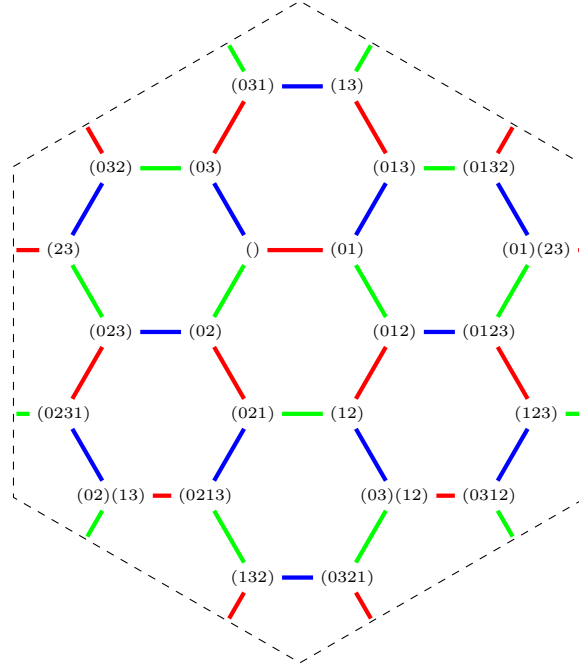


FIGURE 1. $\text{Cay}(G_{\{r,g,b\},1}, \{r,g,b\})$, vertices labeled by permutations of vertices of $T_{3,1}$ with $o = 0$, $o.r = 1$, $o.g = 2$, and $o.b = 3$.

1.1. Comparison to other results. There is a long history in graph theory and in group theory of producing families of bounded valence graphs realizing extreme properties. We focus on girth and diameter in Cayley graphs of finite groups, with fixed number of generators, hence fixed valence. It is known that all possible combinations of valence and girth do indeed occur [12, 9]. There is a tension between girth and diameter in the sense that the naïve estimates (Moore bounds) comparing balls in the graphs to balls in a tree of the same valence show that diameter grows

at least logarithmically and girth grows at most logarithmically in terms of number of vertices of the graph. This makes it a challenge to produce families in which the diameter-to-girth ratio stays bounded.

The property of being an *expander family* implies logarithmic diameter growth, but does not guarantee even that girth is unbounded. In the Cayley graph setting, most constructions of expander families use finite groups of Lie type. The faster growth of alternating groups makes finding expander families of alternating groups harder, although it is possible, by work of Kassabov [14]. Unlike many of the Lie type expander families, ie those of Margulis [17], Lubotzky, Phillips, and Sarnak [16], and Arzhantseva and Biswas [1], Kassabov's construction does not simultaneously produce large girth. See [1, Introduction] for a discussion and extensive bibliography.

There do exist choices of generating sets of alternating and symmetric groups that produce large girth families, due to work of Gamburd, Hoory, Shahshahani, Shalev, and Virág [10], but these are existence results using probabilistic methods.

We compare the behavior of the groups $G_{C,R}$ with these contexts.

Random Cayley graphs. Gamburd et al [10] show that if one chooses a generating set of fixed size uniformly at random in a sufficiently large finite group from within certain families, then almost surely the resulting Cayley graphs have high girth. For the symmetric group their lower bound is roughly the square root of the logarithm of the order of the group [10, Theorem 3], although they conjecture the square root can be removed. These results are probabilistic, not constructive.

In contrast, the Biggs tree construction is entirely explicit: the generators are the colors in C , and their action is described combinatorially via tracks in the colored tree, see Section 3. While we do not know the precise girth, we give an upper bound in Corollary 3.9 that we conjecture is the true girth. This upper bound is exponential in R . Coupled with the main theorem saying that $G_{C,R}$ is the full symmetric or alternating group, this would say that the girth g satisfies $g \asymp \frac{\log(|G_{C,R}|)}{\log(\log(|G_{C,R}|))}$. This falls in between the proven and conjectured asymptotic girth growth rates from Gamburd, et al, and has the advantage of coming from an explicit construction.

Symmetric/alternating expanders. Kassabov [14] constructed bounded valence expander families in alternating and symmetric groups. He writes that the generating sets are 'huge', but with more work they can be replaced by generating sets of size at most 20 [14, Remark 5.1]. The structure of the paper is to prove expansion properties with respect to generating sets that are orders of magnitude larger and then do further work to control the ratio of Kazhdan constants when passing to a suitably well behaved smaller generating set. Although a stated goal of this paper is to construct 'explicit' generating sets, it makes extensive use of the representation theory of the symmetric groups. It also makes no attempt to control girth.

In contrast, the Biggs tree groups can be generated with as few as three generators, and these generators can be written down explicitly as permutations. As far as we know, these are the first explicit families of Cayley graphs of symmetric groups with unbounded girth. We have not considered expansion properties of these Cayley graphs in this paper. One might hope that the explicit nature of the construction will allow for further advances in this direction, without the heavy algebraic requirements.

Diameter-to-girth ratio. Biggs tree groups have been mentioned as a possible candidate for constructing bounded diameter-to-girth ratio families. We show in Proposition 6.1 that this cannot work; the Moore diameter lower bound is on the order of $R(C-1)^R$, while the upper girth bound is on the order of $(C-1)^R$, so the diameter-to-girth ratio tends to infinity with R .

Residual constructions. Wilton [21] proved that if F_r is the free group of rank $r \geq 2$ then for every finitely generated, infinite index subgroup $H \leq F_r$ and every finite set $K \subset F_r - H$ there is a surjection ϕ of F_r onto a finite alternating group in which $\phi(k) \notin \phi(H)$ for all $k \in K$. Taking $H = \{1\}$ and K to be the set of all words of length less than g in F_r with respect to a fixed basis, this implies that the generating set of the alternating group given by the ϕ -image of the basis yields a Cayley graph of girth at least g . The proof is by constructing certain finite graphs covering the r -rose. There are choices involved. In principle, if one makes those choices explicit, then this construction could be used to deduce explicit generating sets of fixed size r yielding Cayley graphs of alternating groups with girth going to infinity. The generators will not be involutions. We have not explored the relation between girth and permutation degree of these constructions, or how that relation compares to the case of Biggs tree groups.

1.2. Notation. For positive functions $f(x)$ and $g(x)$, the notation $f \sim g$ means $\lim_{x \rightarrow \infty} f(x)/g(x) = 1$, while $f = o(g)$ means $\lim_{x \rightarrow \infty} f(x)/g(x) = 0$. The notation $f \preceq g$ means there exists a positive constant A such that $f(x) \leq Ag(x)$ for all sufficiently large x . The notation $f \succeq g$ means $g \preceq f$, and $f \asymp g$ means $f \preceq g$ and $g \preceq f$.

2. PRIMITIVE PERMUTATION GROUPS CONTAINING A CYCLE

The *degree* of a permutation group is the size of the set that it permutes. It is *m-transitive* if it acts transitively on the set of distinct m -tuples. It is *sharply m-transitive* if it is m -transitive and for any given pair of m -tuples there is a unique element taking one to the other. It is *primitive* if it does not preserve any nontrivial partition of the set upon which it acts.

A 2-transitive permutation group of degree $n > 2$ is primitive, since a nontrivial partition of $\{1, \dots, n\}$ contains at least one non-singleton subset $A \supset \{a_1, a_2\}$ and at least one other subset $B \supset \{b\}$, but 2-transitivity gives a permutation σ with $a_1.\sigma = a_1$ and $a_2.\sigma = b$.

Theorem 2.1 (Excerpt² of [13, Theorem 1.2]). *If G is a primitive permutation group of degree n containing a nontrivial cycle g then either G contains $\text{Alt}(n)$ or g fixes $0 \leq k \leq 2$ points and one of the following is true:*

- (0) $k = 0$ and:
 - (a) $C_p \leq G \leq \text{AGL}_1(p)$ for $n = p$ prime.
 - (b) $\text{PGL}_d(q) \leq G \leq \text{P}\Gamma\text{L}_d(q)$ with $d \geq 2$, $n = (q^d - 1)/(q - 1)$, and q a prime power.
 - (c) G is one of $L_2(11)$, M_{11} , or M_{23} with $n = 11, 11, 23$, respectively.
- (1) $k = 1$
- (2) $k = 2$

3. TRACKING THE ACTION

By a *word* we mean either an empty string or a string of letters from C such that no $c \in C$ appears twice in succession. The tree T_C has the property that given a word w with letters in C and a vertex $v \in T_C$ there is a unique edge path starting at v such that the sequence of edge colors along the path is w . We would like a similar property in $T_{C,R}$. A convenient way to arrange this is to augment $T_{C,R}$ by adding mirrored half-edges at the leaves. A *mirrored half-edge* is the quotient of a single edge by an involution that exchanges its vertices and fixes its midpoint. Geometrically, it is isometric to a segment of length $1/2$ with one endpoint identified

²The $k = 1, 2$ cases do not occur in our arguments, so we have omitted their subcases.

with one of the vertices of $T_{C,R}$ and the other designated the ‘mirror’. For each leaf v of $T_{C,R}$ there is a unique incident edge. If the color of that edge is $c_v \in C$ then for each $c \in C - \{c_v\}$ we add a mirrored half-edge colored c to $T_{C,R}$ at v . Formally one could say that $T_{C,R}$ with the added colored mirrored half-edges is the quotient of T_C , with unit edge lengths, by a color preserving isometric group action that is free on vertices but inverts some edges, and the ‘mirrors’ are the images of the edge midpoints with $\mathbb{Z}/2\mathbb{Z}$ stabilizers. Alternatively, one could subdivide the edges of T_C with nontrivial stabilizer first so that the action is free on edges and some vertices have $\mathbb{Z}/2\mathbb{Z}$ stabilizers and the quotient is actually a graph, with some edges of length 1 and some of length $1/2$.

These two points of view give the same geometric space, but we prefer the former and will reserve the word ‘vertex’ for the original vertices of $T_{C,R}$. This allows us to make statements like: for every vertex v of $T_{C,R}$ there is a color preserving isometry between the open ball of radius $1/2$ around v and the open ball of radius $1/2$ about any vertex of T_C . There is a picture in Figure 2. For the purposes of this paper the important point is the following convention: if v is a vertex of $T_{C,R}$ with an incident mirrored half-edge colored c , then the path starting at v corresponding to the word c is the path that travels from v to the mirror of the c -colored mirrored half-edge and then travels back to v . With this convention, for every word w with letters in C and a vertex $v \in T_{C,R}$ there is a unique (edge/mirrored half-edge) path starting at v such that the sequence of colors along the path is w . The path is locally injective except at the points at which it reflects off of a mirror.

A more typical way to define colored edge paths in a colored graph would be to make the path stop and wait at a vertex if there is no incident edge of the correct color. For our application the problem with this approach is that such a path collapses some colored subwords to a vertex, so the path does not remember the length or the full color sequence of the original word. Another approach would be to add colored loops at the leaves of $T_{C,R}$ instead of mirrored half-edges, but this gives those vertices the wrong valence in comparison to T_C and breaks the uniqueness of the path corresponding to a color sequence, since one has a choice of which way to go around such a loop.

Define the *height* of a vertex in $T_{C,R}$ as its distance from the root o . Relative to a vertex v , call the edges leading farther from o the *up edges* and, if $v \neq o$, call the unique edge leading from v closer to o the *down edge*. This description is for convenience of the narration, and is only relative to a fixed vertex; we do not treat $T_{C,R}$ as a directed tree in the graph-theoretic sense. Indeed, since the generators of $G_{C,R}$ are involutions there is no need to record whether traversing an edge colored c in a certain direction corresponds to c or c^{-1} , since these are equivalent in $G_{C,R}$.

The right-action of $G_{C,R}$ on $T_{C,R}$ can be simply visualized as pushing along a path: the element of $G_{C,R}$ represented by a nonempty word $w := c_{i_1} \cdots c_{i_n}$ sends each vertex v of $T_{C,R}$ to the unique vertex at the end of the path starting at v whose color sequence is $c_{i_1} \cdots c_{i_n}$. We will refer to this path as the *track of the action of w on v* . Since we are assuming words are freely reduced, the track of the action of w on v is a path in $T_{C,R}$ that is locally injective except when it reflects off a mirror. Again, see Figure 2.

If γ is a geodesic edge path in $T_{C,R}$ from v to v' whose edges are colored $c_{i_1}, c_{i_2}, \dots, c_{i_n}$ then for $w := c_{i_1} \cdots c_{i_n}$ the track of the action of w on v is exactly the geodesic γ . This shows $G_{C,R} \curvearrowright T_{C,R}$ is transitive; we get an element taking any vertex v to any other vertex v' by pushing v along the unique geodesic from v to v' in $T_{C,R}$.

For $c \in C$, define the *peripheral subgroup* $P_{\neg c}$ of $G_{C,R}$ to be the subgroup generated by all of the generators except c . Define $\partial T_{C,R}$ to be the vertices at height R , and let $\partial_c T_{C,R}$ be the vertices of $\partial T_{C,R}$ whose incident edge is colored c .

Proposition 3.1. *The stabilizer in $G_{C,R}$ of a vertex v is generated by the subgroups $\delta P_{\neg c} \bar{\delta}$ as c ranges over C and δ ranges over geodesics in $T_{C,R}$ from v to a vertex in $\partial_c T_{C,R}$, and where $\bar{\delta}$ denotes the reverse of the path δ . The shortest nonempty words representing elements of $\text{Stab}(v)$ have length $1 + 2d(v, \partial T_{C,R})$, which is the minimal distance from v to a mirror and back.*

Proof. Suppose w is a nonempty word stabilizing $v \in T_{C,R}$. The track of its action is a nontrivial path that begins and ends at the same vertex v and is locally injective except when it reflects off a mirror. Since $T_{C,R}$ is a tree, the path must hit some mirrors, or else it cannot return to v . Thus, the track decomposes as a concatenation of embedded segments $\alpha + \beta_0 + \cdots + \beta_{n-1} + \gamma$ such that:

- α starts at v and ends at a mirror.
- γ starts at a mirror and ends at v .
- If β_i exists then it starts and ends at distinct mirrors.

It follows immediately that $|w| \geq |\alpha| + |\gamma| = 1 + 2d(v, \partial T_{C,R})$, and it remains only to show that w is a product of elements of the desired form.

Suppose first that there are no segments β_i . Then $\gamma = \bar{\alpha}$. Let δ be the initial subsegment of α that includes all of α except the final half-edge. The last edge of δ has some color c , and the last vertex v' is in $\partial_c T_{C,R}$, so $\alpha + \gamma$ follows the unique geodesic δ from v to v' , reflects off one of the mirrors at distance $1/2$ from v' , and then travels along $\bar{\delta}$ back to v . This says w is a word of the form $\delta p \bar{\delta}$ for some single letter $p \in C - \{c\}$. Hence, the group element represented by w is an element of $\delta P_{\neg c} \bar{\delta}$. Now suppose there are some β segments. Consider the geodesic segment β_0 . It contains a unique vertex v' closest to v . Let β'_0 be the geodesic in $T_{C,R}$ consisting of the initial subsegment of β_0 until the point v' , followed by the geodesic from v' to v . Let β''_0 be the geodesic in $T_{C,R}$ that follows the geodesic from v to v' and then the terminal subsegment of β_0 starting from v' . Then $\alpha + \beta'_0$ is as in the previous paragraph, so is represented by a conjugate of a parabolic. Iteratively repeat the argument for $\beta''_0 + \beta_1 + \cdots + \beta_{n-1} + \gamma$. This process realizes w as the free reduction of a product of conjugates of parabolics. $\square$

Corollary 3.2. *In $G_{C,R}$, the stabilizer of the root vertex o is generated by palindromes of length $2R + 1$, and these are the shortest nonempty words representing elements of $\text{Stab}(o)$.*

Proof. The previous proposition shows that $\text{Stab}(o)$ is generated by elements that can be expressed as a word that travels along a geodesic from o to some vertex on $\partial T_{C,R}$, which is distance R away, then uses one letter to reflect off a mirror, and then travels the original path back to o . Such a word has length $2R + 1$ and reads the same forward as backward. $\square$

This implies Biggs's girth bound:

Corollary 3.3. $\text{girth}(\text{Cay}(G_{C,R}, C)) \geq 2R + 1$

Proposition 3.4. *If $C = \{c_1, \dots, c_{|C|}\}$ then $w := \prod_{i=1}^{|C|} c_i$ is an $N_{C,R}$ -cycle.*

Proof. $T_{C,R}$ can be drawn in the plane in such a way that the (half)edges incident to each vertex are colored $c_1, c_2, \dots, c_{|C|}$, when read clockwise starting from the c_1 edge. Then the track of the $w^{N_{C,R}}$ action on o can be imagined as following the boundary of a tubular neighborhood of the embedding of $T_{C,R}$ into the plane, starting and ending at o . See Figure 2.

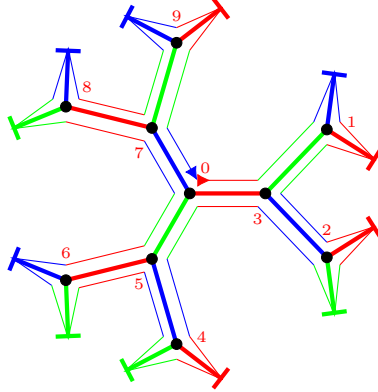


FIGURE 2. $T_{3,2}$ with the track of $(rgb)^{10}$ at o , showing that rgb is a 10-cycle. A vertex is numbered n when that vertex is $o.(rgb)^n$. Mirrors are emphasized by terminal bars.

The track hits each vertex v of $T_{C,R} - \{o\}$ C -many times, once for each possible transition c_i to c_{i+1} or c_C to c_1 . However, only the c_C to c_1 transition at v corresponds to a power of w taking o to v , so $n \mapsto o.w^n$ gives a bijection between numbers $1, \dots, N_{C,R}$ and vertices of $T_{C,R}$. $\square$

This gives us a first girth estimate, as follows, which we will soon improve.

Corollary 3.5. $\text{girth}(\text{Cay}(G_{C,R}, C)) \leq CN_{C,R}$

For any subset $C' \subset C$ the tree $T_{C,R}$ decomposes into a disjoint union of maximal C' -colored subtrees, where each component is a copy of $T_{C',r}$ for some $r \leq R$. The subgroup G' of $G_{C,R}$ generated by C' preserves each such C' -colored component and maps onto the corresponding $G_{C',r}$, simply by tracking the action of words on the component. In general this is a proper quotient, since G' is simultaneously acting on several different C' -colored components, but this observation allows us to describe the cycle structure of elements of G' in terms of induced actions on the C' -colored components. When $C' = 2$ we will also call a C' -colored component a *dichrome arc*. In this case it is an arc with both ends being mirrors of $T_{C,R}$.

Proposition 3.6. $G_{2,R}$ is a dihedral group of order $2(2R+1)$ and $\text{Stab}(o) \cong \mathbb{Z}/2\mathbb{Z}$. The alternating words $c_1c_2\dots$ and $c_2c_1\dots$ of length $2R+1$ represent the same element of $G_{2,R}$ and they act by fixing o and exchanging the two halves of $T_{2,R} - \{o\}$, preserving distance to o .

Proof. By Proposition 3.4, the element $c_1c_2 \in G_{2,R}$ acts as a $(2R+1)$ -cycle on the $2R+1$ vertices of the line graph $T_{2,R}$, and $c_1(c_1c_2)c_1 = c_2c_1 = (c_1c_2)^{-1}$. The peripheral subgroups are isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and $\text{Stab}(o)$ is generated by alternating words $w = c_1c_2\dots$ and $w' = c_2c_1\dots$ of length $2R+1$. But $ww' = (c_1c_2)^{2R+1} = 1$, and $w^2 = (w')^2 = 1$, so $\text{Stab}(o) = \langle w \rangle \cong \mathbb{Z}/2\mathbb{Z}$. Finally, w is an involution in a centerless dihedral group, so $w(c_1c_2)w = (c_1c_2)^{-1}$, so $o.(c_1c_2)^nw = o.w(c_2c_1)^n = o.(c_2c_1)^n$. The distance claim follows from the symmetry exchanging the colors. $\square$

Corollary 3.7. If $2n+1$ is prime relative to $2R+1$ then the alternating words $x := c_1c_2\dots$ and $y := c_2c_1\dots$ of length $2n+1$ generate $G_{2,R}$.

Proof. Since c_1c_2 has order $2R+1$ in $G_{2,R}$ and $2n+1$ is prime relative to $2R+1$, the element $xy = (c_1c_2)^{2n+1}$ also has order $2R+1$. Since $G_{2,R}$ is dihedral of order $2(2R+1)$, it is generated by any involution and element of order $2R+1$. Thus, $\langle x, y \rangle = \langle x, xy \rangle = G_{2,R}$. $\square$

Proposition 3.8. For $w := \prod_{i=1}^C c_i$, let $w^{N_{C,R}/C}$ denote the initial subword of length $N_{C,R}$ of $w^{N_{C,R}}$. Then $w^{N_{C,R}/C}$ acts as a rotation on $T_{C,R}$ that fixes o and takes edges of color c_i to edges of color c_{i+1} .

Proof. For the planar embedding of $T_{C,R}$ as in Proposition 3.4, the word $w^{N_{C,R}/C}$ pushes every vertex of $T_{C,R}$ exactly $1/C$ of the way around the track of $w^{N_{C,R}}$. $\square$

Corollary 3.9. $\text{girth}(\text{Cay}(G_{C,R}, C)) \leq 2N_{C,R}$

Proof. By Proposition 3.8, $(\prod_{i=1}^C c_i)^{N_{C,R}/C}$ is a rotation of $T_{C,R}$. By the same argument, $(\prod_{i=1}^C c_{C+1-i})^{N_{C,R}/C}$ is the opposite rotation. Since $N_{C,R}/C$ is not an integer, the former does not end with c_C , so their product is a freely reduced word of length $2N_{C,R}$ representing the identity permutation. $\square$

Definition 3.10. Define some least common multiple functions:

$$\text{lcm}: \mathbb{N} \rightarrow \mathbb{N} : n \mapsto \text{lcm}\{m \mid 1 \leq m \leq n\}$$

$$\text{oddlcm}: \mathbb{N} \rightarrow \mathbb{N} : n \mapsto \text{lcm}\{2m+1 \mid 1 \leq 2m+1 \leq n\}$$

Proposition 3.11. If $C = 2$ or $R = 1$ then $\text{girth}(\text{Cay}(G_{C,R}, C)) = 4R + 2$. Otherwise, $4R + 3 \leq \text{girth}(\text{Cay}(G_{C,R}, C)) \leq 2\text{oddlcm}(2R + 1)$.

Proof. The girth when $C = 2$ follows from Corollary 3.7, so assume $C > 2$. Let w be a word realizing the girth. If w is dichrome then it acts separately on each of the dichrome arcs of the same two colors. These arcs have $2r + 1$ vertices for each $0 \leq r \leq R$, so Corollary 3.7 gives $|w| = 2\text{oddlcm}(2R + 1) \geq 2(2R + 1)$, with equality only if $R = 1$.

Now suppose w is not dichrome, and decompose it as $w = xyz$ where x is its maximal dichrome prefix, which has length at least 2, and y is the maximal subword of length at most R following x . By hypothesis, y is nonempty, so there is some leaf v such that the geodesic from v to o has color sequence that agrees with y for the length of y . If $|y| < R$ then z is empty. However, if $w = xy$ is a girth witness then $v.w = v$, but $v.x = v$, so $v.xy = v.y$ is the vertex at distance $|y| > 0$ from v along the geodesic from v to o , which is a contradiction. Thus, $|y| = R$ and z is nonempty. The last letter of y is different from the first letter of z , so z leaves o on a different edge than the track of xy from v arrived on. Thus, for $v.xyz = v$, we must have that the track of z from o starts on a different branch than that containing v , travels to a mirror, reflects, then does some more traveling before eventually passing back through o and continuing to v . This gives $|z| \geq 3R + 1$, so $|w| \geq 4R + 3$. $\square$

A similar lower bound argument appears in [20, Section 2.1].

We now compare the upper bounds given by Corollary 3.9 and Proposition 3.11. The maximal power of 2 that occurs in numbers $x \leq n$ is $2^{\lfloor \log_2(n) \rfloor}$, so:

$$\text{oddlcm}(n) = \text{lcm}\{2m+1 \mid 1 \leq 2m+1 \leq n\} = \frac{\text{lcm}(n)}{2^{\lfloor \log_2(n) \rfloor}}$$

A similar argument expresses the second Chebyshev function ψ as $\psi(n) = \log(\text{lcm}(n))$, or, equivalently, $\text{lcm}(n) = e^{\psi(n)}$. The Prime Number Theorem has an equivalent formulation as $\psi(n) \sim n$, and this can be made effective, for example by the following estimate, which will be used in Section 5:

Theorem 3.12 ([6, Theorem 3.3]). For $x \geq 2$:

$$|\psi(x) - x| < \frac{.85x}{\log(x)}$$

Lemma 3.13. Asymptotically in R the bound $2N_{C,R}$ is smaller if $C \leq 8$ and $2\text{oddlcm}(2R + 1)$ is smaller if $C \geq 9$.

Proof.

$$\begin{aligned} \log \left(\frac{2 \operatorname{oddlcm}(2R+1)}{2N_{C,R}} \right) &= \psi(2R+1) - \lfloor \log_2(2R+1) \rfloor \log 2 - \log(N_{C,R}) \\ &= R(2 - \log(C-1)) + o(R) \end{aligned}$$

Whether this goes to $\pm\infty$ as $R \rightarrow \infty$ hinges on whether $C-1 < e^2 \approx 7.4$. $\square$

4. 2-TRANSITIVITY

Proposition 4.1. *If $C > 2$ then $G_{C,R} \curvearrowright T_{C,R}$ is 2-transitive and not sharply 2-transitive.*

Proof. We already know $G_{C,R} \curvearrowright T_{C,R}$ is transitive, so 2-transitivity is equivalent to transitivity of $\operatorname{Stab}(o) \curvearrowright T_{C,R} - \{o\}$.

Let $C = \{c_1, c_2, c_3, \dots\}$. Consider any two colors c_i and c_j . The vertices $o.c_i$ and $o.c_j$ are contained in the (c_i, c_j) -dichrome arc through o . Words in $\langle c_i, c_j \rangle$ act on that dichrome arc as elements of $G_{2,R}$ act on $T_{2,R}$. In particular, by Proposition 3.6, alternating words $c_i c_j \dots$ and $c_j c_i \dots$ of length $2R+1$ fix o and exchange $o.c_i$ and $o.c_j$. Thus, $\operatorname{Stab}(o)$ acts transitively on the set of vertices at height 1.

Next we will show that for any vertex v at height $h > 1$ there exists an element of $\operatorname{Stab}(o)$ that moves v to a vertex at some lower positive height. By induction, every vertex v at positive height is in the same $\operatorname{Stab}(o)$ orbit as some vertex at height 1. This will imply 2-transitivity, since we have already shown that $\operatorname{Stab}(o)$ is transitive on vertices at height 1.

Let v be a vertex at height $h > 1$. Suppose c_1 is the color of the down edge at $v_0 := v$ and c_2 is the color of the down edge at $v_1 := v_0.c_1$. Let $x := c_1 c_3 \dots$ and $y := c_3 c_1 \dots$ be alternating words of length $2R+1$. These are elements of $\operatorname{Stab}(o)$, by Corollary 3.2. The (c_1, c_3) -dichrome arc A_1 through v_0 and v_1 has lowest vertex v_1 and contains $2(R - (h-1)) + 1 = 2(R-h) + 3$ vertices. If $2(R-h) + 3$ is prime relative to $2R+1$ then x and y generate the full dihedral group $G_{2,R-h+1}$ acting on $A_1 \cong T_{2,R-h+1}$, by Corollary 3.7. In particular, there is an element of $\langle x, y \rangle$ that moves v_0 to v_1 , which is at height $h-1 > 0$, and we are done.

Suppose, to the contrary, that $D_1 = \gcd(2(R-h) + 3, 2R+1) \geq 3$. Consider the alternating word $z := c_1 c_3 \dots$ of length $(2(R-h) + 3)(2R+1)/D_1$. Since z is an alternating word whose length is an odd multiple of $2R+1$ we have $z \in \operatorname{Stab}(o)$. On the other hand, its length is an odd multiple of $2(R-h) + 3$, so by Proposition 3.6 it fixes the lowest vertex on A_1 and exchanges the two halves. Thus, $v' := v_0.z = v_0.c_1 c_3$ is in the same $\operatorname{Stab}(o)$ -orbit as v_0 . If $|C| > 3$ then the same construction applies using c_i with $3 < i \leq |C|$ in place of c_3 , so v' is in the same $\operatorname{Stab}(o)$ -orbit as $v_0.c_1 c_i$. The up edges at v_1 are colored by c_1 , c_3 , and c_i for all $3 < i \leq |C|$, so this shows that $\operatorname{Stab}(o)$ acts transitively on the vertices immediately above v_1 . Furthermore, odd numbers with a nontrivial common divisor differ by at least $2D_1$, so $D_1 = \gcd(2(R-h) + 3, 2R+1) \geq 3$ implies $h \geq 4$. Thus, the vertex $v_2 := v_0.c_1 c_2$ is at height at least 2, which implies that there is a down edge at v_2 leading to a vertex v_3 . That down edge cannot be colored c_2 , since v_2 has a c_2 -colored up edge, so without loss of generality it is colored either c_1 or c_3 . In fact, we may assume it is colored c_3 by exchanging vertices v_0 and v' and colors c_1 and c_3 , if necessary.

Let A_2 be the (c_1, c_2) -dichrome arc containing v_0 , v_1 , and v_2 . By construction, the lowest vertex on A_2 is v_2 , since the down edge at v_2 is colored c_3 . The height of v_2 is $h-2$, so A_2 contains $2(R - (h-2)) + 1 = 2(R-h) + 5$ vertices. Repeat the argument from A_1 : if $2(R-h) + 5$ is prime relative to $2R+1$ then the alternating words $c_1 c_2 \dots$ and $c_2 c_1 \dots$ of length $2R+1$ generate a subgroup of $\operatorname{Stab}(o)$ that

acts transitively on A_2 , so v_0 is in the same $\text{Stab}(o)$ -orbit as the vertex v_2 , which is at lower positive height.

The alternative is that $D_2 = \gcd(2R+1, 2(R-h)+5) \geq 3$. As in the previous case, there are elements of $\text{Stab}(o)$ that fix v_2 and exchange the two halves of A_2 . The same holds for every dichrome arc whose lowest vertex is v_2 , so $\text{Stab}(o)$ acts transitively on the upper neighbors of v_2 . Similarly, each such vertex is the lowest vertex on some dichrome arcs containing $2(R-h)+3$ vertices, and there are elements of $\text{Stab}(o)$ reflecting such an arc across its lowest vertex. Conclude that $\text{Stab}(o)$ acts transitively on the set of vertices v in $T_{C,R}$ such that v_2 is the vertex at distance 2 from v on the unique geodesic from v to o . Thus, without changing $\text{Stab}(o)$ -orbits, we may choose the two vertices above v_2 more carefully, as follows. Let c_i be the color of the down edge at v_3 . We know that $c_i \neq c_3$, since $v_3.c_3 = v_2$. Choose j different from i and 3, which is possible since $C \geq 3$.

By the transitivity just established, we may replace v_0 by $v_2.c_j.c_3$ without changing the $\text{Stab}(o)$ -orbit of v_0 or the fact that v_2 lies on the geodesic from v_0 to o . After this replacement, the geodesic from v_0 to v_3 has color sequence $c_3.c_j.c_3$. Moreover, the down edge at v_3 has color c_i , which is different from both c_3 and c_j , so the (c_j, c_3) -dichrome arc through v_0 has lowest vertex v_3 .

In summary, we iteratively find dichrome arcs $A_1, \dots$ such that A_n contains $2(R-h)+2n+1$ vertices, one of which is in the same $\text{Stab}(o)$ -orbit as v , and whose lowest vertex v_n is at height $h-n$. We also have that for all $m < n$:

$$D_m := \gcd(2(R-h)+2m+1, 2R+1) > 1$$

The iteration stops if $D_n = 1$, which happens eventually since $D_{h-1} = 1$. When $D_n = 1$ we use alternating words in the two colors of A_n to construct an element of $\text{Stab}(o)$ that moves v to v_n , which is strictly closer to o . On the other hand, if $D_m > 1$ for all $m \leq n$ then the iterated transitivity argument implies that all vertices v' for which v_n is the vertex at distance n from v' on the geodesic from v' to o are in the same $\text{Stab}(o)$ -orbit. In particular, since $C \geq 3$ we can look at the first two colors c_k and c_ℓ on the geodesic from v_n to o and choose a third color $c_{\ell'}$, distinct from c_k and c_ℓ . Then the $(c_k, c_{\ell'})$ -dichrome arc A_{n+1} through v_n has lowest vertex $v_{n+1} := v_n.c_k$ exactly one step below v_n . Replace v by the unique vertex v' that is on A_{n+1} and such that v_n is distance n from v' along the geodesic from v' to o . Since v' is in the same $\text{Stab}(o)$ orbit as v , and $(n+1)$ -steps above v_{n+1} , this sets up the $n+1$ step of the iteration.

Figure 3 illustrates an example in which several lateral exchanges are necessary before the vertex can be moved down.

Finally, $G_{C,R}$ is not sharply 2-transitive, since the alternating word $c_1 c_2 \dots$ of length $2R+1$ fixes o and every vertex of $\partial_{c_3} T_{C,R}$, but is nontrivial, since it exchanges the two sides of the (c_1, c_2) -dichrome arc through o . $\square$

5. PROOF OF THE MAIN THEOREM

The tree $T_{C,R}$ has $N_{C,R}$ -many vertices and $(N_{C,R}-1)/C = E_{C,R}$ -many edges of each color, for:

$$N_{C,R} = \begin{cases} 2R+1 & \text{if } C = 2 \\ \frac{C(C-1)^{R-2}}{C-2} & \text{if } C > 2 \end{cases} \quad E_{C,R} = \begin{cases} R & \text{if } C = 2 \\ \frac{(C-1)^{R-1}}{C-2} & \text{if } C > 2 \end{cases}$$

Since each generator of $G_{C,R}$ is a product of $E_{C,R}$ -many disjoint transpositions, $G_{C,R} < \text{Alt}(N_{C,R})$ if and only if $E_{C,R}$ is even. But $E_{C,R}$ is odd if C is odd, and if C is even then the parity of $E_{C,R}$ alternates.

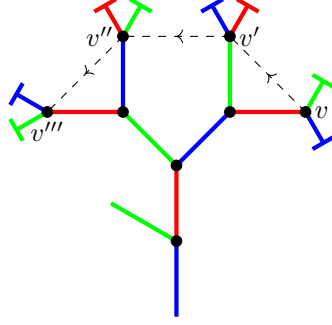


FIGURE 3. In $G_{3,7}$ suppose v is at height 7 on the (r, b) -dichrome arc containing o . All alternating dichrome words of length 15 fix o , but none of them move v , v' , or v'' closer to o , since the dichrome arcs on which these vertices sit have lengths dividing 15. However, an (r, g) alternating word of length 15 swaps v and v' , a (b, g) alternating word of length 15 swaps v' and v'' , and an (r, b) alternating word of length 15 swaps v'' and v''' . Finally, the (r, g) -dichrome arc containing v''' has 7 vertices, prime relative to 15. The alternating word $rg \cdots r$ of length 15 moves v''' closer to o .

For $R = 1$, the generators plus the elements used to permute neighbors of the root in the proof of Proposition 4.1 give all the transpositions, so $G_{C,1} \cong \text{Sym}(C + 1)$, so assume $R > 1$.

By Proposition 3.4, the word $w = \prod_{c \in C} c$ is an $N_{C,R}$ -cycle, fixing 0 points. Apply Theorem 2.1. We are not in Theorem 2.1 case (0c), since 11 and 23 do not occur as possible values of $N_{C,R}$ when $R > 1$. We are not in Theorem 2.1 case (0a), because $\text{AGL}_1(p)$ is sharply 2-transitive, while $G_{C,R}$ is not. This leaves the possibility of Theorem 2.1 case (0b), that $G_{C,R}$ is a subgroup of some projective semi-linear group with the same permutation degree.

One way to rule out the projective case is to produce cycles in $G_{C,R}$ with many fixed points, which we can do in certain cases.

Suppose that $C' \subset C$ is a proper subset containing at least two colors. Consider the word $w = \prod_{c \in C'} c$. By the argument of Proposition 3.4, w acts cyclically on C' -colored components of $T_{C,R}$. These are disjoint subtrees whose sizes are $N_{C',r}$ for each $0 \leq r \leq R$, so the cycle decomposition of w consists of cycles of all of these sizes. Let $m := \prod_{1 \leq r < R} N_{C',r}$. Then w^m acts trivially on all C' -colored subtrees except possibly the unique largest one, the one containing o , of size $N_{C',R}$. If $N_{C',R}$ is prime relative to $N_{C',r}$ for all $r < R$ then $N_{C',R}$ is prime relative to m , so w^m is an $N_{C',R}$ -cycle. Thus, the cycle decomposition of w^m is a single $N_{C',R}$ -cycle, and, since $R > 1$, $N_{C,R} - N_{C',R} > 2$, so the cycle w^m fixes more than 2 vertices. Then Theorem 2.1 says $G_{C,R}$ contains $\text{Alt}(N_{C,R})$. This gives:

- (1) If there is $C' \subset C$ with $2 \leq C' < C$ and $N_{C',R}$ is prime relative to $N_{C',r}$ for all $r < R$ then $G_{C,R}$ contains $\text{Alt}(N_{C,R})$.

This condition is true, in particular, if $N_{2,R} = 2R + 1$ is prime, so for all $C > 2$ there are infinitely many R , including all $R \leq 3$ for which $G_{C,R}$ contains $\text{Alt}(N_{C,R})$.

Now consider $C' = C - 1$, and consider the C' -colored subtrees of $T_{C,R}$. As before, there is a unique largest one, containing o , but this time there is also a unique second largest one, rooted at $o.c$ for the unique $c \in C - C'$. Thus, for w the product of colors of C' , the cycle decomposition of w consists of cycles of sizes $N_{C',r}$ for all $0 \leq r \leq R$, and among these there is a unique $N_{C',R-1}$ cycle. Let $m := \prod_{1 \leq r \leq R, r \neq R-1} N_{C',r}$. If $N_{C-1,R-1}$ is prime relative to all of the other $N_{C-1,r}$

then w^m is an $N_{C-1,R-1}$ cycle, and $N_{C,R} - N_{C-1,R-1} > 2$, so Theorem 2.1 says $G_{C,R}$ contains $\text{Alt}(N_{C,R})$.

- (2) If $N_{C-1,R-1}$ is prime relative to $N_{C-1,r}$ for all $r < R-1$ and for $r = R$, then $G_{C,R}$ contains $\text{Alt}(N_{C,R})$.

Table 1 gives the prime factorizations of $N_{C,R}$. By (1), if an entry $N_{C,R}$ is prime relative to every entry above it in its column then $G_{D,R}$ contains $\text{Alt}(N_{D,R})$ for all $D > C$, and the entries $N_{D,R}$ to the right of $N_{C,R}$ in its row are colored violet in Table 1. For example all of the row $R = 2$ for $C \geq 3$ is violet because $N_{2,2} = 5$ is prime relative to $N_{2,1} = 3$.

Condition (2) applies to say that $G_{C,R}$ contains $\text{Alt}(N_{C,R})$ whenever we look at the entry $N_{C-1,R-1}$ above and to the left of $N_{C,R}$, and if that entry is prime relative to the entry $N_{C-1,R}$ immediately below it and to every entry $N_{C-1,r}$ above it in its column. We color such entries $N_{C,R}$ in Table 1 blue if they are not already violet. For example $G_{3,4}$ contains $\text{Alt}(N_{3,4})$, and entry $R = 4, C = 3$ is blue, because 7 is prime relative to 9 and to both of 3 and 5.

$R \setminus C$	2	3	4	5	6	7
1	3	2^2	5	$2 \cdot 3$	7	2^3
2	5	$2 \cdot 5$	17	$2 \cdot 13$	37	$2 \cdot 5^2$
3	7	$2 \cdot 11$	53	$2 \cdot 53$	$11 \cdot 17$	$2 \cdot 151$
4	3^2	$2 \cdot 23$	$7 \cdot 23$	$2 \cdot 3 \cdot 71$	937	$2 \cdot 907$
5	11	$2 \cdot 47$	$5 \cdot 97$	$2 \cdot 853$	$43 \cdot 109$	$2 \cdot 5443$
6	13	$2 \cdot 5 \cdot 19$	$31 \cdot 47$	$2 \cdot 3413$	$23 \cdot 1019$	$2 \cdot 11 \cdot 2969$
7	$3 \cdot 5$	$2 \cdot 191$	4373	$2 \cdot 3^2 \cdot 37 \cdot 41$	$7 \cdot 16741$	$2 \cdot 5 \cdot 39191$
8	17	$2 \cdot 383$	13121	$2 \cdot 13 \cdot 4201$	$11 \cdot 53267$	$2 \cdot 673 \cdot 1747$
9	19	$2 \cdot 13 \cdot 59$	$5 \cdot 7873$	$2 \cdot 218453$	2929687	$2 \cdot 7054387$
10	$3 \cdot 7$	$2 \cdot 5 \cdot 307$	$7 \cdot 16871$	$2 \cdot 3 \cdot 291271$	1487 · 9851	$2 \cdot 13 \cdot 3255871$
11	23	$2 \cdot 37 \cdot 83$	$19 \cdot 29 \cdot 643$	$2 \cdot 3495253$	$59 \cdot 349 \cdot 3557$	$2 \cdot 67 \cdot 631 \cdot 6007$
12	5^2	$2 \cdot 6143$	1062881	$2 \cdot 13981013$	577 · 634681	$2 \cdot 5 \cdot 163 \cdot 271 \cdot 6899$

TABLE 1. Prime factorizations of $N_{C,R}$.

For the remaining 6 entries in the table with $C > 2$, which are colored red, we checked by brute force computation that there is no $d \geq 2$ and prime power q such that $(q^d - 1)/(q - 1) = N_{C,R}$. However, we cannot always exclude such an equality. For example, when C is a prime power, $q = C^2$, and $R = d = 2$ there is a coincidental equality $N_{C,R} = (q^d - 1)/(q - 1)$. These particular examples do not bother us, since we have already established the $R = 2$ case of the theorem, but we do not know if there are any other such coincidences for larger R , so we turn to another approach for ruling out $G_{C,R} \leq \text{PGL}_d(q)$. The idea is to demonstrate an element of $G_{C,R}$ whose order is too high relative to $N_{C,R}$ for it to occur in a projective semi-linear group of the same permutation degree.

For an element g in a group G , let $\text{ord}(g)$ be the order of g in G , and let $\text{meo}(G) := \max\{\text{ord}(g) \mid g \in G\}$. The group $\text{PGL}_d(q)$ is a semi-direct product of $\text{PGL}_d(q)$ with the Galois group of the field extension of $\mathbb{F}_q$ over $\mathbb{F}_p$. It embeds into $\text{Aut}(\text{PSL}_d(q))$.

If $G_{C,R} \leq \text{PGL}_d(q)$ then $N_{C,R} = \frac{q^d - 1}{q - 1}$. The exceptional cases $(d, q) = (2, 4), (3, 2)$ cannot occur in this setting, since in those cases we do not have $N_{C,R} = \frac{q^d - 1}{q - 1}$ for $C > 2$ and $R > 2$. Outside the exceptional cases, the values of $\text{meo}(\text{PGL}_d(q))$ and $\text{meo}(\text{Aut}(\text{PSL}_d(q)))$ agree, and are equal to $\frac{q^d - 1}{q - 1}$ [11, Corollary 2.7 and Theorem 2.16/Table 3], so:

$$(3) \quad \text{meo}(\text{PGL}_d(q)) = \frac{q^d - 1}{q - 1} = N_{C,R}$$

This gives a bound for $\log(\text{meo}(\text{PGL}_d(q)))$, when $\text{PGL}_d(q)$ is of degree $N_{C,R}$, that asymptotically grows linearly in R with slope $\log(C - 1)$.

The goal for the remainder of this section is to identify an element of $G_{C,R}$ whose order is greater than the maximum given in (3), which is impossible if $G_{C,R} \leq \text{PTL}_d(q)$, so we have $G_{C,R} \not\leq \text{PTL}_d(q)$.

First specialize to $C = 3$. Consider the element $c_1 c_2 \in G_{3,R}$. As in the dichrome case of the proof of Proposition 3.11, the element $c_1 c_2$ has order $\text{oddlcm}(2R+1)$. Thus Theorem 3.12 gives:

$$\begin{aligned} \log(\text{ord}(c_1 c_2)) &= \log\left(\frac{\exp(\psi(2R+1))}{2^{\lfloor \log_2(2R+1) \rfloor}}\right) \\ &> \log\left(\frac{\exp((1 - \frac{.85}{\log(2R+1)})(2R+1))}{2R+1}\right) \end{aligned}$$

Define:

$$F(R) := \log\left(\frac{\exp((1 - \frac{.85}{\log(2R+1)})(2R+1))}{2R+1}\right) - \log(N_{3,R})$$

Then $G_{3,R} \not\leq \text{PTL}_d(q)$ if $F(R) > 0$. The $\log(\text{ord}(c_1 c_2))$ term grows linearly in R with slope 2, which is higher than the asymptotic slope of the $\log(\text{meo}(\text{PTL}_d(q)))$ term, so $F(R) > 0$ for sufficiently large R . It is a calculus exercise to check that $F(R)$ is increasing for $R \geq 1$. Figure 4 shows that F already becomes positive at $R = 5$, so we conclude that it stays positive for all $R \geq 5$. Thus, $G_{3,R}$ is not a subgroup of $\text{PTL}_d(q)$.

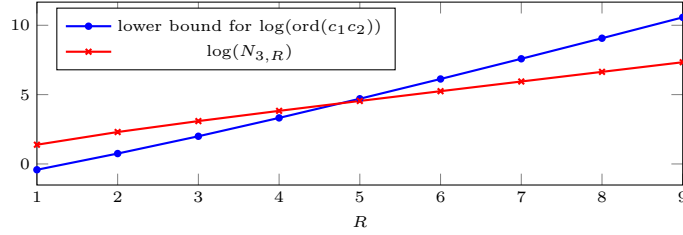


FIGURE 4. The lower bound for $\log(\text{meo}(G_{3,R}))$ is already greater than the upper bound for $\log(\text{meo}(\text{PTL}_d(q)))$ when $R = 5$.

We do not push this computation to higher numbers of colors. Indeed, for $C \geq 9$ the asymptotic slope $\log(C-1)$ of $\log(\text{meo}(\text{PTL}_d(q)))$ is greater than the asymptotic slope 2 of $\log(\text{ord}(c_1 c_2))$. However, now that we know that $G_{3,R}$ is the symmetric group, it turns out that the choice of the explicit element $c_1 c_2$ was not a particularly good one in terms of establishing a lower bound on $\text{meo}(G_{C,R})$. The maximum element order of $\text{Sym}(n)$ is equal to the maximum, over partitions of n , of the least common multiple of the numbers in the partition. The function³ $\Lambda: n \mapsto \text{meo}(\text{Sym}(n))$ is called the *Landau function*, after Landau, who proved [15] that $\log(\Lambda(n)) \sim \sqrt{n \log(n)}$. There are effective lower bounds for sufficiently large inputs, cf [18, (1.6)]:

$$(4) \quad \log(\Lambda(n)) \geq \sqrt{n \log(n)} \quad \text{for } n \geq 906$$

There are also several thousand precomputed values for small n [19, A000793].

For $C > 3$, we induct on the number of colors: Suppose that for $2 < C' < C$ we know that $G_{C',R}$ contains $\text{Alt}(N_{C',R})$ for all sufficiently large R . Let H be the subgroup of $G_{C,R}$ generated by a set C' consisting of $C-1$ colors. Then H surjects onto $G_{C',R}$ and $G_{C',R-1}$ via the action on the largest and second largest

³The Landau function is usually denoted $g(n)$ in the literature.

C' -colored components of $T_{C,R}$. If C' or R is odd let $R' := R$, and if both are even let $R' := R - 1$, so that by the induction hypothesis, once R is large enough, $G_{C',R'} \cong \text{Sym}(N_{C',R'})$. Take an element h of maximal order $\Lambda(N_{C',R'})$ in $G_{C',R'}$, and lift it to an element $\tilde{h} \in H$. Then $\text{ord}(\tilde{h})$ is a multiple of $\text{ord}(h)$, so:

$$\log(\text{meo}(G_{C,R})) \geq \log(\text{ord}(\tilde{h})) \geq \log(\Lambda(N_{C',R'}))$$

Thus, when R is large enough that (4) applies, $\log(\text{meo}(G_{C,R}))$ is bounded below by an exponential function of R :

$$\log(\text{meo}(G_{C,R})) \geq \sqrt{N_{C',R'} \log(N_{C',R'})} \geq (\sqrt{C-2})^{R-1}$$

On the other hand, $\log(\text{meo}(\text{PTL}_d(q))) \sim R \log(C-1)$ is linear. Conclude that for sufficiently large R , the group $G_{C,R}$ contains elements whose orders are too high in relation to $N_{C,R}$ to belong to a projective semi-linear group of the same permutation degree.

To close the remaining gap and show that the desired conclusion holds for all R , it is convenient to handle the case of large C first. For $C > 10$ and $R \geq 3$ we have $N_{C-1,R'} > 906$, so the bound of (4) gives:

$$Q(C, R) := \frac{\sqrt{N_{C-1,R'} \log(N_{C-1,R'})}}{\log(N_{C,R})} \leq \frac{\log(\Lambda(N_{C',R'}))}{\log(N_{C,R})} \leq \frac{\log(\text{meo}(G_{C,R}))}{\log(\text{meo}(\text{PTL}_d(q)))}$$

Then $N_{C-1,R'} > (C-2)^{R-1}$ and $\log(N_{C,R}) < (R+1) \log(C-1)$, so:

$$Q(C, R) > \frac{(C-2)^{(R-1)/2} \sqrt{(R-1) \log(C-2)}}{(R+1) \log(C-1)}$$

The right-hand side is increasing in R , increasing in C for $C \geq 11$, and greater than 1 at $R = 3$ and $C = 11$. Thus, $Q(C, R) > 1$ for all $C \geq 11$ and $R \geq 3$, and we conclude that $G_{C,R} \not\leq \text{PTL}_d(q)$ for these ranges of C and R .

For $3 < C \leq 10$, the above argument applies once R is large enough that $N_{C-1,R'} \geq 906$, so that the estimate (4) applies. In each of these cases this happens before $R = 13$, so since we already know $G_{C,R} \not\leq \text{PTL}_d(q)$ for $R < 13$, this completes the proof.

Figures 5 and 6 use precomputed values of Λ to illustrate that $\log(\Lambda(N_{C',R'}))$ is already much larger than $\log(N_{C,R})$ for small values of R when $C = 4, 5$.

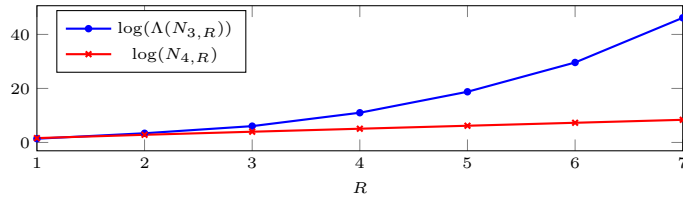


FIGURE 5. The lower bound for $\log(\text{meo}(G_{4,R}))$ is already greater than the upper bound for $\log(\text{meo}(\text{PTL}_d(q)))$ when $R \geq 2$.

6. FURTHER REMARKS

We conclude with a few remarks on some well-known open questions surrounding Biggs tree groups for $C > 2$.

Question 1. *What is the precise girth of $\text{Cay}(G_{C,R}, C)$?*

There is quite a wide gap between the linear, in terms of R , girth lower bound and near exponential upper bounds given by Corollary 3.9 and Proposition 3.11.

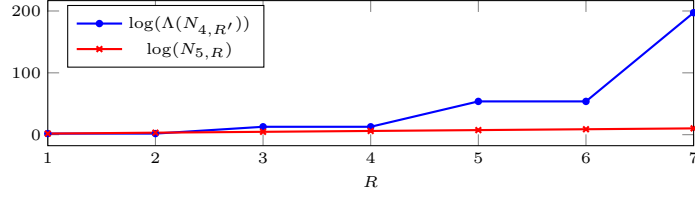


FIGURE 6. The lower bound for $\log(\text{meo}(G_{5,R}))$ is already greater than the upper bound for $\log(\text{meo}(\text{PTL}_d(q)))$ when $R \geq 3$.

Question 2. What is the diameter of $\text{Cay}(G_{C,R}, C)$?

Using the main theorem: $\text{Cay}(G_{C,R}, C)$ has at least $N_{C,R}/2$ vertices. If D is the diameter, the ball of radius D contains all of these. A ball of radius r in a C -regular graph contains at most the number $N_{C,r}$ of vertices in a ball in the tree of the same valence. For fixed C and growing R , this gives a lower *Moore bound* on D by a constant multiple of $R(C-1)^R$.

Proposition 6.1. For any $C > 2$ and sequence of radii $R \rightarrow \infty$ the Cayley graphs $\text{Cay}(G_{C,R}, C)$ do not form a bounded diameter-to-girth ratio family.

Proof. The family $\text{Cay}(G_{C,R}, C)$ consists of C -regular graphs with girth going to infinity, but since the order of the groups grows so fast, even with the most optimistic bounds on diameter and girth the ratio cannot stay bounded:

$$\frac{\text{diam}(\text{Cay}(G_{C,R}, C))}{\text{girth}(\text{Cay}(G_{C,R}, C))} \succeq \frac{R(C-1)^R}{(C-1)^R} \xrightarrow{R \rightarrow \infty} \infty$$

Here the estimates on the diameter and girth are from the Moore bound and Corollary 3.9, respectively. $\square$

Question 3. Is there a $C > 2$ such that for some sequence of R 's the family $\text{Cay}(G_{C,R}, C)$ is an expander family?

The exponential diameter estimate above is in terms of R , not $|G_{C,R}|$, so it is not necessarily an obstacle to being an expander family.

Finally, one might wonder about other Biggs groups; that is, what is the permutation group defined by some other colored graph, instead of a finite symmetric tree? In that case, we could again add mirrored half-edges at every vertex as necessary so that every vertex has exactly one incident (half)edge of each color. Then the procedure of tracking the action of words works just as in the tree case, and this point of view may be useful in the analysis of the resulting group. However, both the proof of 2-transitivity and the production of cycles depended heavily on the symmetry of $T_{C,R}$. Moreover, every finite group generated by involutions occurs as a Biggs group: take the colored Schreier graph of its right regular action with respect to those generating involutions—this gives the original group as the story of the picture. Thus, rather than seeking uniform results for arbitrary pictures, a more natural approach may be to identify special families of colored graphs whose structure makes computation of the story tractable.

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